# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **How do Organisms Reproduce?** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Starting with one yeast cell, after 4 successive buddings of each cell that each give one new cell (doubling), the number is 1

**(a)** 8 **(b)** 32 **(c)** 4 **(d)** 16

**2.** Which of these is NOT a correct pair? 1

**(a)** Plasmodium - budding **(b)** Rhizopus - spores

**(c)** Spirogyra - fragmentation **(d)** Planaria - regeneration

**3.** Assertion (A): Banana plants bear fruits through seeds. Reason (R): Banana is propagated by vegetative methods. 1

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**4.** A pea flower in a bag, before it opens, still sets seeds. This shows 1

**(a)** cross-pollination is the only way

**(b)** self-pollination can occur within a flower

**(c)** gametes are not needed

**(d)** pollination is not needed

**5.** In a population of bacteria, a heat-resistant variant survives a heat wave. This demonstrates 1

**(a)** the survival value of variation **(b)** budding

**(c)** the failure of reproduction **(d)** importance of regeneration

**6.** Assertion (A): The uterine lining thickens every month. Reason (R): It prepares to receive a fertilised egg. 1

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**7.** Identify the true statement about pollen tube. 1

**(a)** It is formed in the anther before landing

**(b)** It grows out of the ovary

**(c)** It forms the fruit

**(d)** It grows towards the ovule through the style

**8.** A plant growing from a seed is genetically different from the parent, but a plant from a leaf of Bryophyllum is 1

**(a)** a hybrid **(b)** completely different

**(c)** the product of two parents **(d)** nearly identical to the parent

**9.** **Assertion (A):** Sperms are produced in the testes. 1
**Reason (R):** The testes are located inside the abdominal cavity of a male.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** Fertilisation in human beings takes place in the uterus. 1
**Reason (R):** The zygote gets implanted in the lining of the uterus.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** If the vas deferens is blocked, what will be the effect on a man's fertility? State why. 2

**12.** How many seeds would a fruit have if its ovary had three ovules, of which only one was fertilised? Give a reason. 2

**13.** Why do plants produced from tissues of the same parent have the same traits but those formed from seeds do not? 2

**14.** Why does an embryo not require a digestive system in the uterus? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** How does the size of a gamete relate to its function in humans? 3

**16.** A newborn is found to have genetic characters from both parents. Explain with reference to gametes. 3

**17.** Compare the way in which a Hydra and a flowering plant produce offspring. 3

OR

Explain why a group of plants developed from stem cuttings is more prone to a disease outbreak than one raised from seeds. 3

**18.** Why is the fruit of a plant often juicy or winged? 3

OR

How does the placenta ensure that the mother's and the embryo's blood do not mix but exchange occurs? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A gardener, Ramu, grows new rose plants from cuttings of stems and sprouts new potato plants from the buds (eyes) of old potatoes. In his bakery, he uses yeast to make bread, and notices small outgrowths on yeast cells under the microscope. 5
(i) What is the method of growing plants from vegetative parts called?
(ii) Name the part of the potato that gives rise to new plants.
(iii) Name the mode of reproduction in yeast.
(iv) Why is vegetative propagation preferred for growing banana and rose plants?

**20.** Explain why plants grown by vegetative propagation are early bearers but genetically less variable. Discuss one risk to farmers. 5

OR

Compare binary fission, multiple fission, budding and fragmentation in a table and mention one organism for each. 5

## END OF QUESTION PAPER — Total: 40 Marks